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Risk Factors for Pituitary Macrotumor in Dogs With Hypercortisolism

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ABSTRACT

Background: Diagnosing pituitary macrotumor in dogs with hypercortisolism requires advanced imaging, which is not feasible for every case. Identifying risk factors that can guide the decision to pursue imaging would be valuable.

Objective: Determine clinical and physical examination findings that are associated with an increased likelihood of pituitary macrotumors in dogs with hypercortisolism.

Animals: A total of 130 dogs diagnosed with hypercortisolism.

Methods: Retrospective study to analyze medical records for data on age, sex, breed, clinical signs, physical findings, endocrine test results, imaging results, diagnosis, and treatment. Logistic regression was used to identify risk factors for pituitary macrotumor, defined by a pituitary height/brain area ratio ≥ 0.4.

Results: Risk factors for pituitary macrotumor included diagnosis of hypercortisolism at ≤ 10.9 years of age (odds ratio [OR], 0.718; $p < 0.05$), French Bulldog breed (OR, 21.0; $p < 0.01$), and presence of neurologic signs (OR, 10.9; $p < 0.001$).

Conclusion and Clinical Importance: Advanced pituitary imaging should be recommended in dogs with hypercortisolism ≤ 10.9 years of age, French Bulldogs, and those presenting with neurological signs because these factors significantly increase the likelihood of pituitary macrotumor in dogs with hypercortisolism.

1 | Introduction

Hypercortisolism is a common endocrine disorder in dogs that markedly affects quality of life and is manifested by clinical signs such as polyuria, polydipsia, increased appetite, abdominal distension, endocrine alopecia, skin abnormalities (e.g., comedones, telangiectasia, decreased elasticity of the skin, pyoderma, and calcification of the skin), and muscle wasting [1]. Approximately 85% of hypercortisolism cases in dogs are attributed to pituitary tumors, known as pituitary-dependent hypercortisolism (PDH). These tumors are categorized as pituitary microtumors or macrotumors based on their size or the pituitary height/brain area (*P*/*B*) ratio [2].

Pituitary macrotumors not only cause hypercortisolism but also can result in neurologic signs because of brain parenchyma compression from the mass effect [1]. Furthermore, the presence of a macrotumor influences treatment approach, success rate, and prognosis [3–5]. Trilostane is currently the most commonly used medical treatment for hypercortisolism. However, it may suppress adrenal–pituitary axis feedback, potentially accelerating pituitary tumor growth and exacerbating the mass effect

Abbreviations: ACTH, adrenocorticotropic hormone; CT, computed tomography; IQR, interquartile range; MRI, magnetic resonance imaging; OR, odds ratio; *P*/*B* ratio, pituitary height/brain area ratio; PDH, pituitary-dependent hyperadrenocorticism.

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[6, 7]. Early diagnosis is therefore essential for identifying pituitary macrotumor [4]. However, the use of pituitary imaging tests for all dogs with PDH remains challenging because of high cost, anesthesia risk, and limited accessibility.

Efforts have been made to identify risk factors for pituitary macrotumors using common clinical tests. A previous study found that body temperature ($\leq 38.3°C$) and heart rate (≤ 84 beat per minute [bpm]) were associated with increased risk [8]. However, that study was limited to these two variables and did not include multivariable analysis of other factors. Additionally, although pathologic studies suggest that pituitary macroadenomas may be more common in brachycephalic breeds, no significant differences were identified [9]. Research on risk factors for pituitary macrotumors remains limited, but identifying such factors by evaluating routine clinical tests and signalment could provide valuable guidance for advanced imaging decisions. We aimed to identify risk factors for pituitary macrotumors in dogs with hypercortisolism by evaluating signalment, physical examination findings, and clinical signs.

2 | Materials and Methods

2.1 | Dogs

Our retrospective study analyzed clinical cases of hypercortisolism in dogs. We reviewed the electronic medical records and imaging data of dogs presented to Gifu University Animal Medical Center from December 2012 to December 2022. Dogs diagnosed with hypercortisolism based on clinical signs, along with adrenal imaging and at least one dynamic endocrine test (adrenocorticotropic hormone [ACTH] stimulation test or low-dose dexamethasone suppression test) conducted at our hospital or a referral hospital, were included. Cases diagnosed with PDH based on bilateral adrenal enlargement without shape irregularity, and with pituitary imaging (computed tomography [CT] or magnetic resonance imaging [MRI]), performed at the time of diagnosis or later, were selected for inclusion. Cases with unclear diagnoses or insufficient data for analysis were excluded. The study was approved by the local ethics committee for animal clinical research (E24008), and written consent for the use of clinical data in the research was obtained from all owners.

2.2 | Data Collection

Data on age, sex, breed, medical history, clinical signs, physical examination findings, endocrine tests, imaging studies (ultrasonography, CT, or MRI), diagnosis, and medications at the time of diagnosis were collected from the electronic medical records and imaging data of the included cases. Adrenal gland size was evaluated by calculating the average thickness of cranial and caudal poles of both the right and left adrenal glands. Pituitary height and brain cross-sectional area were measured from CT or MRI scans, and the cases were classified into two groups based on the P/B ratio (pituitary height [mm]/brain area [mm^2] $\times 100$). The P/B ratio was measured three times on different days by a single researcher, and the mean value was used in the study. Cases with a P/B ratio $\geq 0.40 \times 10^{-2}\,mm^{-1}$ were assigned to the pituitary macrotumor group, whereas those with a P/B ratio $< 0.40 \times 10^{-2}\,mm^{-1}$ were placed in the pituitary microtumor group [10].

2.3 | Statistical Analysis

The collected data were managed using electronic spreadsheets (Microsoft Excel) and analyzed using statistical analysis software (JMP Pro 16.2.0; SAS Institute Inc., Cary, NC). The univariate and multivariate analyses were performed by EZR software version 1.41 (Saitama Medical Center, Jichi Medical University, Saitama, Japan). The Shapiro–Wilk test was used to assess data normality, with results presented as median and interquartile range (IQR). A Welch's *t*-test or the Wilcoxon rank sum test was used to compare continuous variables between the two groups. Fisher's exact test was applied to compare proportions between the groups. Variables that showed significant differences in the univariate analysis were included in the multivariate logistic regression analysis. The forward–backward stepwise selection method was used to determine which variables to include. Receiver operating characteristic (ROC) curve analysis was used to assess the diagnostic utility of age at diagnosis to differentiate between dogs with pituitary macroadenoma and dogs with pituitary microadenoma. Sensitivity and specificity were calculated, and the optimal cutoff with the highest Youden index ([sensitivity + specificity] − 1) was selected. A p-value of < 0.05 was considered significant for all analyses.

3 | Results

3.1 | Dogs

Between December 2012 and December 2022, 238 dogs with hypercortisolism were presented to our hospital. Of these, 29 dogs (12%) were diagnosed with adrenal-dependent hypercortisolism, and 209 dogs (88%) were diagnosed with PDH. Of the 209 PDH cases, 131 dogs (63%) underwent pituitary evaluation using CT or MRI. Specifically, CT alone was performed in 116 dogs, MRI alone in 2 dogs, and both CT and MRI in 13 dogs. One dog was excluded because of missing hormonal testing data, leaving 130 dogs in the study. Based on imaging results, 54 dogs (41.5%) were classified as having pituitary macrotumors, and 76 dogs (58.5%) as having pituitary microtumors. Among dogs with pituitary macrotumors, 43 (80%) had already been diagnosed with hypercortisolism before pituitary evaluation, or the diagnosis and imaging were performed on the same day. The median time from diagnosis to imaging was 41 days (interquartile range [IQR], 0–264). In contrast, among the dogs with pituitary microtumors, 60 (79%) had already been diagnosed with hypercortisolism before pituitary evaluation, or the diagnosis and imaging were performed on the same day. For this group, the median time from diagnosis to imaging was 11 days (IQR, 0–80).

3.2 | Signalment

The signalments of dogs in the pituitary macrotumor and pituitary microtumor groups are summarized in Table 1. Dogs in the pituitary macrotumor group were significantly younger at the time of hypercortisolism diagnosis compared with those

TABLE 1 | Summary of treatment status, duration, and signalment of dogs.

Variables	Macrotumor ($n = 54$)	Microtumor ($n = 76$)	p
Age (year), median [IQR]			
	9.9 [8.5–11]	11.7 [9.4–13]	< 0.0001
Sex, n (%)			
Intact male	12 (22)	7 (9)	0.15
Neutered male	19 (35)	24 (32)	
Intact female	6 (11)	13 (17)	
Spayed female	17 (32)	32 (42)	
Body weight (kg), median [IQR]			
Untreated	6.9 [3.7–12]	7.3 [4.6–10]	0.85
Treated	8.6 [5.4–11]	7.4 [4.7–9.3]	0.21
Therapeutic interventions at diagnosis, n (%)			
Treated	34 (63)	32 (42)	< 0.05
Not treated	20 (37)	44 (58)	
Period, median [IQR]			
Treatment–imaging test (days)	175 [42–372]	88 [34–405]	< 0.01

Abbreviation: IQR, interquartile range.

in the pituitary microtumor group (9.9 years [IQR, 8.5–10.8] vs. 11.7 years [IQR, 9.4–13.1], $p < 0.0001$). No significant differences in sex and neuter status distribution were identified between the two groups ($p = 0.15$). Among dogs that had not received treatment for hypercortisolism, no significant difference in median body weight was found between the two groups (6.9 kg [IQR, 3.7–11.9] vs. 7.3 kg [IQR, 4.6–10.2], $p = 0.85$). Similarly, in dogs receiving treatment for hypercortisolism, no significant difference in median body weight was found between the two groups (8.6 kg [IQR, 5.4–10.5] vs. 7.4 kg [IQR, 4.7–9.3], $p = 0.21$).

The included cases consisted of 24 purebred and mixed breed dogs, with Miniature Dachshunds ($n = 29$) and French Bulldogs ($n = 19$) used for comparison among the purebreds (Table 2). French Bulldogs were significantly more common in the pituitary macrotumor group ($p < 0.005$). However, no significant difference in macrotumor prevalence was found in Miniature Dachshunds.

3.3 | Clinical Signs

In the pituitary macrotumor group ($n = 54$), the following clinical signs were observed: polyuria and polydipsia in 38 dogs (70%), polyphagia in 7 dogs (13%), abdominal distension in 19 dogs (35%), alopecia in 16 dogs (30%), skin abnormalities (comedones, telangiectasia, decreased elasticity of the skin, pyoderma or calcification of the skin) in 16 dogs (30%), weakness in 12 dogs (22%), and neurologic signs in 37 dogs (69%). Conversely, in the pituitary microtumor group ($n = 76$), the following clinical signs were observed: polyuria and polydipsia in 46 dogs (61%), polyphagia in 5 dogs (7%), abdominal distension in 28 dogs (37%), alopecia in 18 dogs (24%), skin abnormalities in 13 dogs (17%), weakness in 8 dogs (11%), and neurologic signs in 19 dogs (25%). Neurologic signs, especially depressed mentation and seizures, were significantly more common in the pituitary macrotumor group ($p < 0.001$; Tables 3, 4).

Among dogs that were already receiving treatment for hypercortisolism at the time of pituitary imaging evaluation, 29 dogs (85%) in the pituitary macrotumor group and 19 dogs (59%) in the pituitary microtumor group showed at least one clinical sign, such as polyuria and polydipsia, polyphagia, abdominal distension, alopecia, skin abnormalities, or weakness. A significantly higher proportion of dogs in the pituitary macrotumor group continued to exhibit clinical signs after treatment compared with those in the pituitary microtumor group ($p < 0.05$).

3.4 | Physical Examination

The results for body temperature, heart rate, and respiratory rate in dogs from the pituitary macrotumor and pituitary microtumor groups are shown in Table 5. Among dogs with untreated hypercortisolism at the time of physical examination, body temperature in the pituitary macrotumor group was significantly lower than in the pituitary microtumor group (median [IQR], 38.5°C [38.1–38.8] vs. 38.8°C [38.5–39.2], $p < 0.01$). However, in both groups, the body temperature of most dogs remained within the normal range. No significant difference in heart rate was found between the two groups (median [IQR], 115 bpm [100–132] vs. 120 bpm [110–148], $p = 0.1$). Because some dogs were panting, respiratory rate was difficult to assess accurately. Respiratory rates were categorized using a cutoff of ≥ 40 breaths/min, but no significant difference was found between the two groups ($p = 1.0$).

TABLE 2 | Distribution of breeds among dogs with macrotumor and microtumor.

		Macrotumor ($n = 54$)	Microtumor ($n = 76$)	Odds of Macroadenoma			
Breed	**Total**	***n* (%)**	***n* (%)**	**OR**	**95% CI**	***p***	**Variable *p***
Crossbred	10	6 (11.1)	4 (5.3)		Baseline		0.0002
Miniature Dachshund	29	10 (18.5)	19 (25.0)	0.682	0.288,1.61		
French Bulldog	19	15 (27.8)	4 (5.3)	6.92	2.15,22.3	0.001	
Toy Poodle	11	4 (7.4)	7 (9.2)	0.467	0.06,3.56		
Chihuahua	10	0 (0)	10 (13.2)		NE		
Welsh Corgi	7	2 (3.7)	5 (6.6)	0.389	0.03,4.80		
Miniature Schnauzer	6	0 (0)	6 (7.9)		NE		
Yorkshire Terrier	6	1 (1.9)	5 (6.6)		NE		
Shih Tzu	5	2 (3.7)	3 (3.9)	0.583	0.04,7.66		
Shiba Ken	5	2 (3.7)	3 (3.9)	1.56	0.165,14.7		
Boston Terrier	4	3 (5.6)	1 (1.3)	7.0	0.501,97.8		
Jack Russell Terrier	3	2 (3.7)	1 (1.3)	1.17	0.07,18.3		
Beagle	2	1 (1.9)	1 (1.3)		NE		
Pomeranian	2	1 (1.9)	1 (1.3)		NE		
American Cocker Spaniel	1	0 (0)	1 (1.3)		NE		
English Springer Spaniel	1	0 (0)	1 (1.3)		NE		
Golden Retriever	1	1 (1.9)	0 (0)		NE		
Kai Ken	1	0 (0)	1 (1.3)		NE		
Maltese	1	1 (1.9)	0 (0)		NE		
Newfoundland	1	0 (0)	1 (1.3)		NE		
Papillon	1	0 (0)	1 (1.3)		NE		
Pug	1	1 (1.9)	0 (0)		NE		
Siberian Husky	1	1 (1.9)	0 (0)		NE		
Shetland Sheepdog	1	1 (1.9)	0 (0)		NE		
Standard Poodle	1	0 (0)	1 (1.3)		NE		

Abbreviations: CI, confidence interval; NE, not estimable; OR, odds ratio.

TABLE 3 | Clinical signs observed in dogs with macrotumor and microtumor.

	Macrotumor ($n = 54$)		Microtumor ($n = 76$)		
Variables	**Untreated**	**Treated**	**Untreated**	**Treated**	***p***
Polyuria and polydipsia	13	15	29	17	0.27
Polyphagia	2	5	2	3	0.23
Abdominal enlargement	9	10	17	11	1.0
Alopecia	6	10	12	16	0.54
Skin abnormalities	5	11	9	4	0.13
Weakness	7	5	4	4	0.09
Neurologic signs	14	23	9	10	< 0.0001*
At least one sign associated with hypercortisolism	17	29	37	19	< 0.05**

*Untreated and treated macrotumor versus untreated and treated microtumor.
**Treated macrotumor versus treated microtumor.

TABLE 4 | Neurological signs observed in dogs with macrotumor and microtumor.

	Macrotumor	Microtumor	Total	
Neurological signs	***n* = 54**	***n* = 76**	***n* = 130**	***p***
Staggering	8	7	15	1.0
Depressed mental state	13	1	14	< 0.001
Seizure	11	1	12	< 0.01
Wandering	5	2	7	1.0
Collapse	5	1	6	1.0
Tremor	6	0	6	0.07
Circling	5	1	6	1.0
Cognitive deficits	5	1	6	1.0
Head press	4	0	4	0.44
Facial paralysis	3	1	4	1.0
Dysstasia	2	2	4	1.0
Paresis	1	3	4	1.0
Blindness	1	2	3	1.0
Personality change	3	0	3	1.0
Dysmetria	1	0	1	1.0
Horner's syndrome	0	1	1	1.0

TABLE 5 | Physical examination results of untreated dogs.

	Macrotumor	Microtumor	
Variables	**(*n* = 20)**	**(*n* = 44)**	***p***
Temperature (°C), median [IQR]			
	38.5 [38.1–38.8]	38.8 [38.5–39.2]	< 0.01
Heart rate (bpm), median [IQR]			
	115 [100–132]	120 [110–148]	0.10
Respiratory rate (breaths/min), *n* (%)			
≥ 40 breaths/min	6 (35)	11 (32)	1.0
< 40 breaths/min	11 (65)	23 (68)	

Abbreviation: IQR, interquartile range.

3.5 | Imaging Tests

No significant difference in adrenal gland thickness, as measured by ultrasonography, was found between the pituitary macrotumor and pituitary microtumor groups (median [IQR], 8.5 mm [7.4–10] vs. 8.0 mm [7.1–10], $p = 0.34$). To evaluate the impact of treatment, dogs treated with *o*,*p*′-DDD were excluded, and the remaining dogs in both groups were compared within treated and untreated subgroups. No significant difference in mean adrenal gland thickness was observed between treated and untreated dogs in the pituitary macrotumor group (median [IQR], 8.9 mm [7.5–10.6] vs. 8.0 mm [7.1–9.0], $p = 0.07$) or in the pituitary microtumor group (median [IQR], 7.8 mm [6.9–10] vs. 8.0 mm [7.2–9.8], $p = 0.96$).

Median pituitary height was 12.1 mm (IQR, 8.0–15) in the pituitary macrotumor group and 2.7 mm (IQR, 2.2–3.6) in the pituitary microtumor group. The median *P*/*B* ratio was $0.81 \times 10^{-2}\,mm^{-1}$ (IQR, 0.62–1.02) in the pituitary macrotumor group and $0.19 \times 10^{-2}\,mm^{-1}$ (IQR, 0.16–0.27) in the pituitary microtumor group. No significant difference in the *P*/*B* ratio was observed when comparing treated and untreated subgroups within either the pituitary macrotumor group (median [IQR], $0.80 \times 10^{-2}\,mm^{-1}$ [0.62–1.0] vs. $0.83 \times 10^{-2}\,mm^{-1}$ [0.60–1.1], $p = 0.50$) or the pituitary

microtumor group (median [IQR], $0.21 \times 10^{-2}\,mm^{-1}$ [0.16–0.29] vs. $0.18 \times 10^{-2}\,mm^{-1}$ [0.15–0.25], $p = 0.13$).

3.6 | Treatment and Time From Diagnosis to Imaging

At the time of pituitary evaluation, 34 dogs in the pituitary macrotumor group and 32 dogs in the pituitary microtumor group already were receiving medical treatment for hypercortisolism, with a significantly higher proportion in the pituitary macrotumor group ($p < 0.05$). Trilostane (mean ± SD dose, 2.2 ± 1.2 mg/kg/day) was used for all treated cases, except for one dog in the pituitary microtumor group that was treated with *o,p′*-DDD. For dogs that had already undergone treatment for hypercortisolism, median treatment duration at the time of pituitary imaging was 175 days (IQR, 42–372) in the pituitary macrotumor group and 88 days (IQR, 34–405) in the pituitary microtumor group ($p < 0.01$; Table 1).

3.7 | Multivariate Logistic Regression Analysis

Significant predictors for pituitary macrotumor included duration from treatment start to imaging, age at diagnosis, French Bulldog breed, body temperature, and neurologic signs. Using a stepwise selection procedure, no factors met the exclusion criterion, and consequently all were included in the multivariate analysis (Table 6). The analysis found that age at diagnosis of hypercortisolism (odds ratio [OR], 0.718; $p < 0.05$), French Bulldog breed (OR, 21.0; $p < 0.01$), and neurologic signs (OR, 10.9; $p < 0.001$) were significant risk factors for the pituitary macrotumor.

The area under the ROC curve (95% CI) for age at diagnosis of hypercortisolism was 0.701 (0.611, 0.791). The optimal cutoff value was 10.9 years, with a sensitivity of 78% and a specificity of 59% at that cutoff.

4 | Discussion

We identified neurologic signs (especially depressed mentation and seizures), French Bulldog breed, and age ≤ 10.9 years at hypercortisolism diagnosis as risk factors for pituitary macrotumor. Although pituitary evaluation is recommended for all dogs with PDH, owners of dogs with these risk factors are strongly advised to have advanced imaging performed to assess for potential pituitary macrotumor.

TABLE 6 | Multivariate logistic regression analysis results.

Variables	OR	95% CI		VIF	*p*
		Low	High		
Days from treatment initiation to image test	1.0	0.999	1	1.0	0.29
Age at diagnosis of hypercortisolism	0.718	0.522	0.967	1.0	0.04
French bulldog	21.0	2.97	148	1.2	0.002
Body temperature	1.2	0.73	1.98	1.1	0.47
Neurologic signs	10.9	2.77	43.0	1.2	0.001

Abbreviations: CI, confidence interval; OR, odds ratio; VIF, variance inflation factor.

In our study, approximately 40% of dogs with PDH were diagnosed with a pituitary macrotumor, which is consistent with previous reports that show a 10%–60% occurrence of pituitary macrotumors in dogs with PDH [9, 11, 12]. To more accurately assess risk factors for pituitary macrotumors from a pathophysiological standpoint [10], we employed a criterion for defining pituitary macrotumor in our study (*P*/*B* ratio $\geq 0.40 \times 10^{-2}\,mm^{-1}$) that differed from that used in earlier reports (pituitary height > 10 mm or *P*/*B* ratio $> 0.31 \times 10^{-2}\,mm^{-1}$). In addition, selection bias in the inclusion of dogs with PDH may account for differences in the observed incidence of pituitary macrotumor compared to previous studies. For example, in our study, we included the performance of advanced imaging as an inclusion criterion. However, advanced imaging may have been performed because of neurologic signs caused by pituitary macrotumors.

Neurologic signs of an enlarged pituitary tumor are common in dogs with PDH. In our study, approximately 70% of dogs with a pituitary macrotumor exhibited neurologic signs. Additionally, among the neurologic signs, depressed mentation and seizures were significantly more frequent in dogs with pituitary macrotumor. The tumor can extend to the pituitary stalk, compressing the hypothalamus, third ventricle, and thalamus, which leads to neurologic signs [13]. Because many of these signs arise from the tumor's space-occupying effect rather than hypercortisolism itself, they can be indicators of an enlarged pituitary tumor in dogs with PDH [14]. Thus, it is expected that dogs showing neurologic signs are more likely to have a pituitary macrotumor.

Of the 19 French Bulldogs included in our study, 15 were diagnosed with pituitary macrotumor, suggesting that French Bulldogs with PDH may have risk factors for pituitary enlargement. Previous studies also have noted that French Bulldogs are more prone to primary intracranial tumors [15, 16]. Additionally, research has indicated that brachycephalic breeds might be more susceptible to developing pituitary macroadenomas [9]. This finding supports the results of our study, in which a high prevalence of pituitary macrotumors was observed not only in French Bulldogs but also in Boston Terriers. The specific reasons for this finding are unclear. In humans, most pituitary tumors are sporadic, but some are associated with genetic mutations [17]. Although genetic mutations that could contribute to pituitary tumor enlargement exist in brachycephalic dogs, no such association has been identified to date. Another possible explanation is that the cranial conformation in brachycephalic breeds may restrict compensatory mechanisms against alterations in intracranial pressure, thereby predisposing these animals to the earlier manifestation of clinical signs associated with intracranial tumors [18]. If this consideration encourages more proactive use of pituitary imaging in such breeds, it may in turn contribute to the higher reported frequency of clinical diagnoses of pituitary macrotumors.

We have shown that the diagnosis of hypercortisolism at ≤ 10.9 years old is associated with a higher incidence of pituitary macrotumor. Similarly, previous research found that

dogs with pituitary macrotumor were younger than those with pituitary microtumor (9 vs. 11 years) [8]. Although the reason for this observation is unclear, one possibility is that dogs with functional pituitary macrotumors show clinical signs because of the mass effect of the tumor, prompting owners to recognize the condition as more severe and seek an earlier diagnosis. Furthermore, in humans, a missense mutation in *CABLES1* has been reported to cause young-onset Cushing's disease with pituitary macrotumor [19]. However, the role of genetic factors in the development of pituitary macrotumors in dogs remains unknown.

Previous studies have reported decreased body temperature and heart rate as clinical findings associated with pituitary macrotumor in dogs [8]. In humans, large pituitary adenomas may compress the hypothalamus, leading to autonomic dysfunction and consequent hypothermia [20, 21]. Likewise, increased intracranial pressure from tumor mass effect can trigger Cushing's reflex, resulting in secondary bradycardia [22]. However, our multivariate analysis did not identify a significant association between hypothermia or bradycardia and pituitary macrotumors. The dogs in our study had less severe tumor enlargement compared with those in previous studies (median P/B ratio, 0.81×10^{-2} vs. $1.0 \times 10^{-2}\,\mathrm{mm}^{-1}$). This difference may be associated with a lesser severity of increased intracranial pressure in the dogs of our study compared with those of earlier studies.

In our study, treatment with trilostane did not have a significant effect on the P/B ratio. In humans with PDH, Nelson's syndrome, which involves rapid growth of ACTH-secreting pituitary tumors after bilateral adrenalectomy, occurs in approximately 20% of cases [23]. This phenomenon also may apply to dogs, and trilostane-induced loss of negative feedback on the hypothalamic–pituitary–adrenal axis could cause Nelson's syndrome. However, many dogs in our study were poorly controlled and showed clinical signs of hypercortisolism at enrollment. Therefore, trilostane may have had only a minor effect on the negative feedback mechanism in the dogs included in our study. We could not obtain imaging data before and after trilostane treatment. Additional prospective studies are necessary to conclude whether trilostane induces chemical Nelson's syndrome in dogs.

Our study had some limitations. First, because of the small sample size, risk assessment only could be performed for a limited number of breeds, and some breeds with predispositions may not have been examined. Second, because our study was retrospective in nature, not all dogs were evaluated using standardized methods or procedures. Therefore, some evaluations may have been incomplete, and certain findings may have been overlooked. In particular, neurologic signs induced by pituitary tumors can be subtle and may have been masked by other, more prominent, clinical signs of hypercortisolism. Third, we only assessed the size of the pituitary gland at a single point in time and did not evaluate changes over time. As a result, early-stage cases of pituitary macrotumor may have been misclassified in the pituitary microtumor group. Fourth, comorbidities that could have affected clinical signs, body temperature, heart rate, and respiratory rate were not considered, potentially confounding the results (Table S1). Further research is needed in untreated dogs with PDH and no comorbidities to provide a more accurate assessment of risk factors for pituitary macrotumor.

Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the Gifu University ethics committee for animal clinical research (E24008). Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. E. N. Behrend, "Canine Hyperadrenocorticism," in *Canine and Feline Endocrinology*, 4th ed., ed. E. C. Feldman, R. W. Nelson, C. E. Reusch, et al. (Elsevier, 2015), 377–451.

2. H. S. Kooistra, G. Voorhout, J. A. Mol, and A. Rijnberk, "Correlation Between Impairment of Glucocorticoid Feedback and the Size of the Pituitary Gland in Dogs With Pituitary-Dependent Hyperadrenocorticism," *Journal of Endocrinology* 152 (1997): 387–394.

3. M. S. Kent, D. Bommarito, E. Feldman, and A. P. Theon, "Survival, Neurologic Response, and Prognostic Factors in Dogs With Pituitary Masses Treated With Radiation Therapy and Untreated Dogs," *Journal of Veterinary Internal Medicine* 21 (2007): 1027–1033.

4. B. P. Meij, G. Voorhout, T. S. G. A. M. Van Den Ingh, et al., "Results of Transsphenoidal Hypophysectomy in 52 Dogs With Pituitary-Dependent Hyperadrenocorticism," *Veterinary Surgery* 27 (1998): 246–261.

5. J. M. Hanson, M. M. Van Hoofd, G. Voorhout, et al., "Efficacy of Transsphenoidal Hypophysectomy in Treatment of Dogs With Pituitary-Dependent Hyperadrenocorticism," *Journal of Veterinary Internal Medicine* 19 (2005): 687–694.

6. T. Teshima, Y. Hara, S. Takekoshi, et al., "Trilostane-Induced Inhibition of Cortisol Secretion Results in Reduced Negative Feedback at the Hypothalamic-Pituitary Axis," *Domestic Animal Endocrinology* 36 (2009): 32–44.

7. E. H. Bertoy, E. C. Feldman, R. W. Nelson, A. B. Dublin, M. H. Reid, and M. S. Feldman, "One-Year Follow-Up Evaluation of Magnetic Resonance Imaging of the Brain in Dogs With Pituitary-Dependent Hyperadrenocorticism," *American Journal of Veterinary Research* 208 (1996): 1268–1273.

8. G. Benchekroun, L. Desquilbet, M. E. Herrtage, N. D. Jeffery, D. Rosenberg, and N. Granger, "Dogs With Macroadenomas Have Lower Body Temperature and Heart Rate Than Dogs With Microadenomas," *Veterinary Journal* 227 (2017): 42–45.

9. M. A. Miller, D. S. Bruyette, J. C. Scott-Moncrieff, et al., "Histopathologic Findings in Canine Pituitary Glands," *Veterinary Pathology* 55 (2018): 871–879.

10. N. Granger, P. De Fornel, P. Devauchelle, et al., "Plasma Pro-Opiomelanocortin, Pro-Adrenocorticotropin Hormone, and Pituitary Adenoma Size in Dogs With Cushing's Disease," *Journal of Veterinary Internal Medicine* 19 (2005): 23–28.

11. A. Marcinowska, J. Warland, M. Brearley, and J. Dobson, "Comparison of Two Coarse Fractionated Radiation Protocols for the Management of Canine Pituitary Macrotumor: An Observational Study of 24 Dogs," *Veterinary Radiology and Ultrasound* 56 (2015): 554–562.

12. E. N. Behrend, H. S. Kooistra, R. Nelson, C. E. Reusch, and J. C. Scott-Moncrieff, "Diagnosis of Spontaneous Canine Hyperadrenocorticism: 2012 ACVIM Consensus Statement (Small Animal)," *Journal of Veterinary Internal Medicine* 27 (2013): 1292–1304.

13. E. N. Behrend and R. J. Kemppainen, "Diagnosis of Canine Hyperadrenocorticism," *Veterinary Clinics of North America. Small Animal Practice* 31 (2001): 985–1003.

14. M. Menchetti, L. De Risio, G. Galli, et al., "Neurological Abnormalities in 97 Dogs With Detectable Pituitary Masses," *Veterinary Quarterly* 39 (2019): 57–64.

15. R. B. Song, C. H. Vite, C. W. Bradley, and J. R. Cross, "Postmortem Evaluation of 435 Cases of Intracranial Neoplasia in Dogs and Relationship of Neoplasm With Breed, Age, and Body Weight," *Journal of Veterinary Internal Medicine* 27 (2013): 1143–1152.

16. J. M. Snyder, F. S. Shofer, T. J. Van Winkle, et al., "Canine Intracranial Primary Neoplasia: 173 Cases (1986–2003)," *Journal of Veterinary Internal Medicine* 20 (2006): 669–675.

17. S. L. Asa, S. Ezzat, and O. Mete, "Clinical Integration and Application of the 2022 WHO Pituitary Tumor Classification," *Neuro-Oncology Advances* 7 (2025): i10–i16.

18. M. W. Fox, "Developmental Abnormalities of the Canine Skull," *Canadian Journal of Comparative Medicine and Veterinary Science* 27 (1963): 219–222.

19. L. C. Hernández-Ramírez, R. Gam, N. Valdés, et al., "Loss-Of-Function Mutations in the CABLES1 Gene Are a Novel Cause of Cushing's Disease," *Endocrine-Related Cancer* 24 (2017): 379–392.

20. M. Shibata, "Hyperthermia in Brain Hemorrhage," *Medical Hypotheses* 50 (1998): 185–190.

21. P. W. Carmel, "Vegetative Dysfunctions of the Hypothalamus," *Acta Neurochirurgica* 75 (1985): 113–121.

22. W. Fitch, D. G. McDowall, N. P. Keaney, et al., "Systemic Vascular Responses to Increased Intracranial Pressure," *Journal of Neurology, Neurosurgery, and Psychiatry* 40 (1977): 843–852.

23. J. W. Miller and L. Crapo, "The Medical Treatment of Cushing's Syndrome," *Endocrine Reviews* 14 (1993): 443–458.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Comorbidities of dogs with PDH.